Predictive Analytics of Climate Stress Impact on the Italian Mediterranean Buffalo

Physiological and Productive Responses

Prof. Luca Barbierato

Prof. Matteo Santinello

Mohammad Sheikh

Politecnico di Torino

1859

Department of Control and Computer Engineering

Dipartimento di Automatica e Informatica

Milk predictions of buffalo

....

Precision livestock farming (PLF)

Why: increasing demand for animal-derived foods requires enhancement in increasing efficiency of farms and production.

How: PLF represents a valuable strategy to tackle challenges in this area. PLF, uses digital technologies like sensors, IoT, AI, and data analytics to monitor and manage individual animals.

Gap: While PLF technologies have been extensively adopted for **pigs**, **poultry**, and **dairy cattle**, there is a scarcity of research regarding **buffalo-specific** PLF applications.

G. Neglia, M. Santinello, *et al* "Precision livestock farming in buffalo species: a sustainable approach for the future," in *Revista Científica de la Facultad de Ciencias Veterinarias*, 2023, doi: 10.52973/rcfcv-wbc019.

Heat Stress is affecting …

↓ milk production

↓ feed intake

↓ performance.

↑ Diseases

↑ Sweating

↑ Blood flow to skin

↑ heart and respiration rate

Bernabucci, *et al*. "Metabolic and Hormonal Acclimation to Heat Stress in Domesticated Ruminants." in *Animal* 2010, 4, 1167–1183, doi : 10.1017/S175173111000090X

Temperature-Humidity Index (THI)

A temperature-humidity index (THI) is a single value representing the combined effects of air temperature and humidity associated with the level of thermal stress.

Normally THI is Based on Temperature in Fahrenheit.

$$THI = T_f - (0.55 - 0.55 \times RH) \times (T_f - 58)$$

$$THI = 0.8 \times T_c + RH \times (T_c - 14.4) + 46.4$$

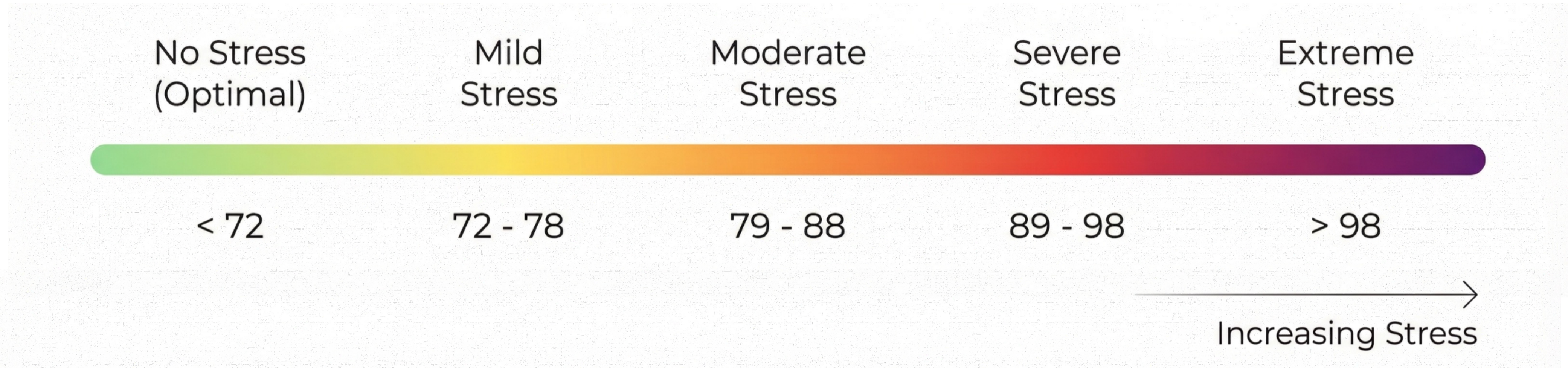
T_f: Temperature in degrees Fahrenheit.

T_c: Temperature in degrees Celsius.

RH: Relative Humidity (decimal).

Bohmanova, J, *et al*. “Temperature-humidity indices as indicators of milk production losses due to heat stress.” in *Journal of Dairy Science*, 90(4), 1947–1956. doi: 10.3168/jds.2006-513

THI refers to be an important KPI in milk production.



V. L. de Araujo *et al*, "Responses of dairy buffalo to heat stress conditions and mitigation strategies: A review," in *Animals*, vol. 14, no. 5, Art. no. 731, Feb. 2024, doi: 10.3390/ani13071260

S. Dash, *et al*, "Effect of heat stress on reproductive performances of dairy cattle and buffaloes: A review," in *Veterinary World*, vol. 9, no. 3, pp. 235–244, Mar. 2016, doi: 10.14202/vetworld.2016.235-244.

Studies on Milk Prediction

- An effective way to incorporate temperature–humidity index to study effect of heat stress on milk yield by an XGBoost machine learning model ,by Hassan, 2025

- Development of individual models for predicting cow milk production for real-time monitoring, by Song, 2025

- A machine learning framework to predict the next month's daily milk yield, milk composition and milking frequency for cows in a robotic dairy farm, by Ji,2022

mohammad.sheikh@studenti.polito.it

Effect of HS on cow milk yield by Hassan, 2025

Case Study: 10 commercial dairy farms in Australia involving 3,369 lactating cows (primarily Holstein Friesian).

Data:

- **Timeframe:** from early 2019 to mid-2023
- **Milk Data:** Daily total yield, milking time, duration, etc.
- **Animal Info:** National ID, birth date, calving date, etc.
- **Weather Data:** a station **<65 km** from the farm.

Novelty: Replaced standard daily THI with **THI Composite**

THI Composite is average of 5-day window of THI= $\frac{(THI_{n-2}+THI_{n-1}+THI_n+THI_{n+1}+THI_{n+2})}{5}$

Methodology: Developed an **eXtreme Gradient Boosting (XGBoost)** model to predict daily milk

Performance & Metrics:

- $R^2 = 0.73$
- Lin's Concordance Correlation Coefficient (LCCC) = 0.846.
- **THI composite** improved prediction accuracy by up to **21%**

M. F. Hasan, *et al* "An effective way to incorporate temperature–humidity index to study effect of heat stress on milk yield by an XGBoost machine learning model," in *Journal of Dairy Science*, vol. 108, no. 12, 2025, doi: 10.3168/jds.2025-27193.

individual models for predicting cow milk production in Korea, by Song,2025

Case Study: 4 commercial dairy farms in South Korea involving 727 cows

Data:

- **Timeframe:** from Jan 2016 to Dec 2018 (3 years)
- **Milk Data:** Daily production volume logged individually via RFID tags.
- **Animal Info:** Parity and Days in Milk (DIM).
- **Weather Data:** no weather data used.

Novelty:

- Developed **individual-based** models**.**
- Modified **Wilmink** Model used for prediction

Methodology: Combined **Simple Exponential Smoothing** with **Modified Wilmink regression** to predict daily milk

Performance & Metrics:

- $R^2 = 0.875$
- Root Mean Squared Error (RMSE) = 2.192

J.-W. Song, *et al* "Development of individual models for predicting cow milk production for real-time monitoring," in *Computers and Electronics in Agriculture*, vol. 228, Art. no. 109698, Jan. 2025, doi: 10.1016/j.compag.2024.109698.

predict the next month's daily cow milk yield in Australia by Ji,2022

Case Study: 1 commercial robotic dairy farm in Australia involving 80 lactating cows.

Data:

- **Timeframe:** from 2013 to 2018 (5 years).
- **Milk Data:** Daily yield (DMY), composition (fat, protein), milking frequency, milk temperature.
- **Animal Info:** Cow ID, age, body mass, days in milk (DIM), dry matter intake (DMI).
- **Weather Data:** a station **~2 km** from the farm

Novelty:

- **individual cow** prediction over a **28-day**

Methodology: Developed **eXtreme Gradient Boosting (XGBoost)** regression and classification models to predict future production

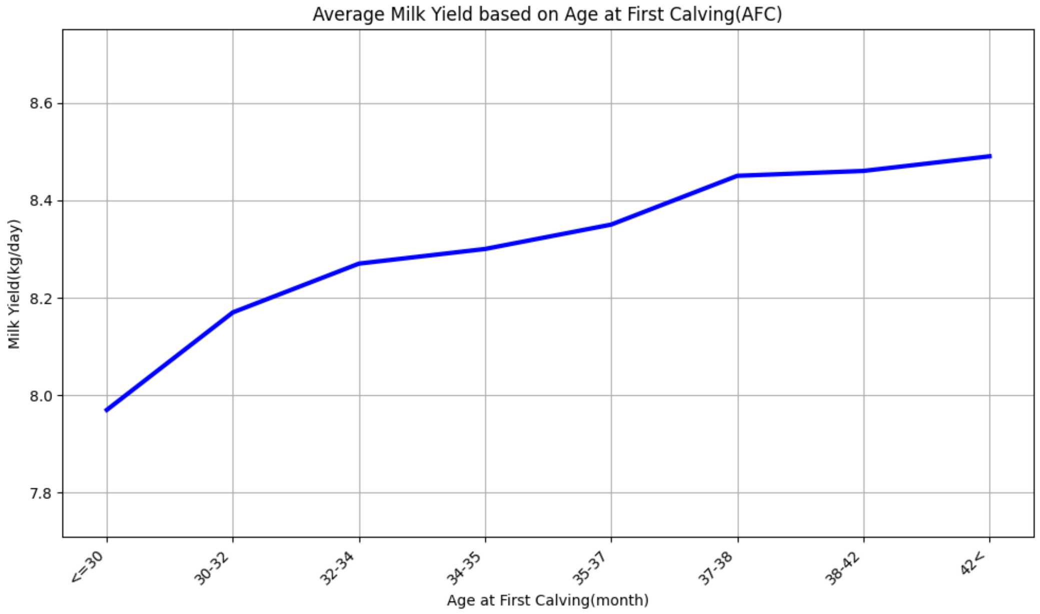
Performance & Metrics:

- mean accuracy for daily milk yield = 91.92 %
- $R^2 > 0.9$

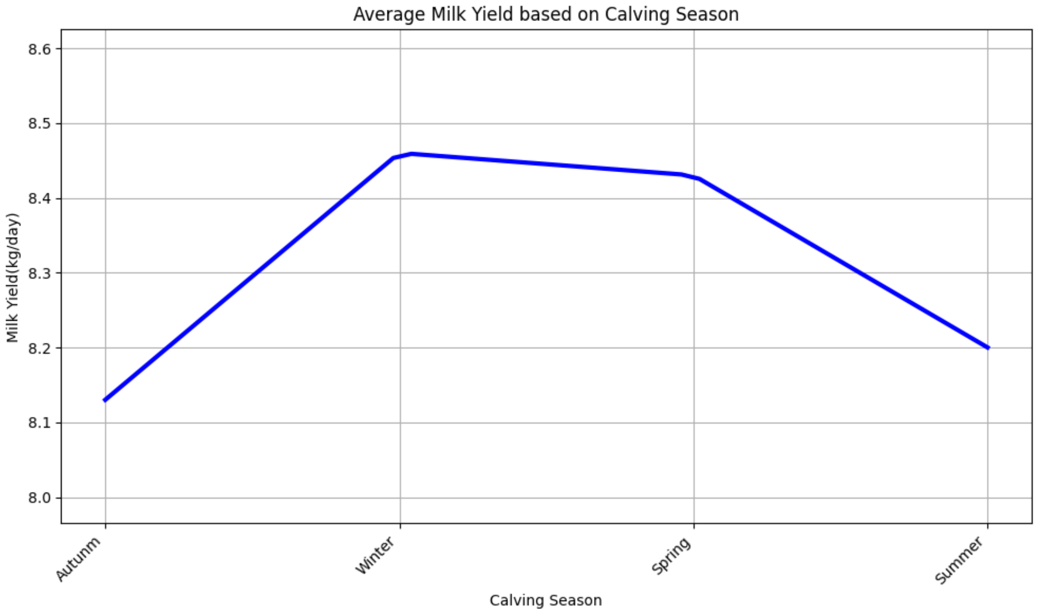
B. Ji *et al.,* "A machine learning framework to predict the next month's daily milk yield, milk composition and milking frequency for cows in a robotic dairy farm," in *Biosyst. Eng.*, vol. 216, pp. 186–197, Apr. 2022, doi: 10.1016/j.biosystemseng.2022.02.013.

Effect of Calving on Milk Yield

A higher age at first calving is associated with an increased average daily milk yield.



Average daily milk yield is significantly influenced by the calving season, with peak production observed during the winter and spring months.

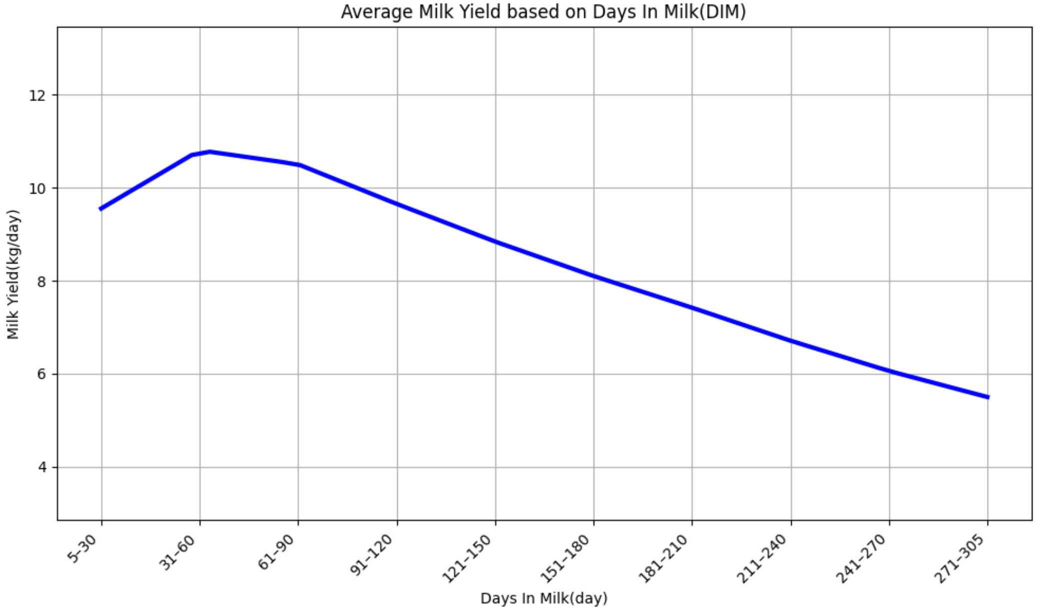


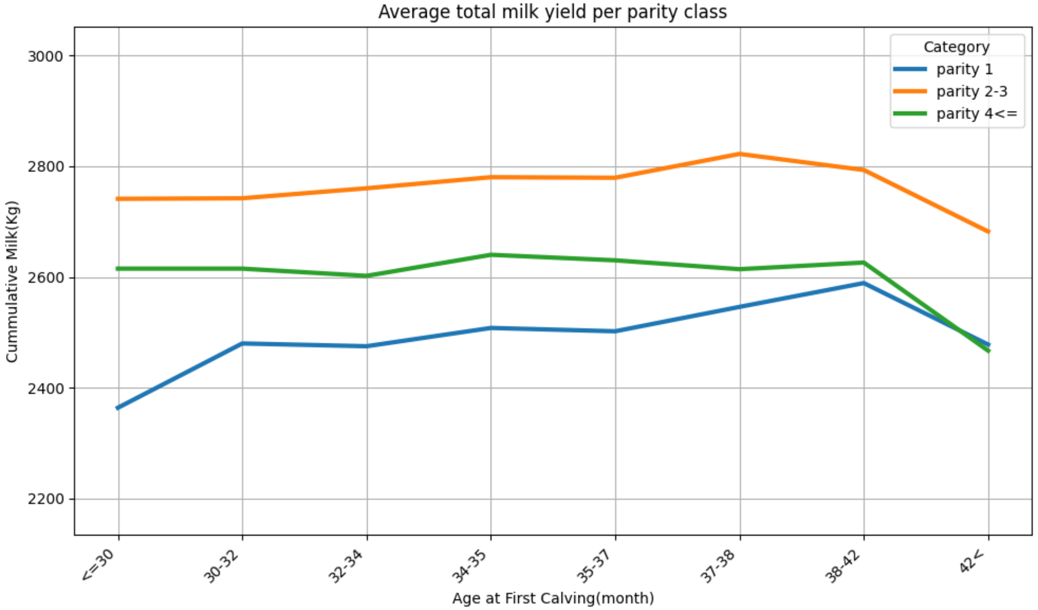
M. Santinello, G. Neglia, *et al.*, "The role of age at first calving in shaping production and reproductive outcomes in Italian buffaloes," in *J. Dairy Sci.*, vol. 108, no. 7, pp. 7235–7247, Jul. 2025, doi: 10.3168/jds.2025-26369.

Effect of Days in Milk and Parity on Milk Yield

Daily milk output peaks roughly one to two months after calving and then steadily decreases.

Cumulative milk yield is highest in their second and third parities, regardless of their age at first calving.





M. Santinello, G. Neglia *et al.*, "The cost of being early or late: Biological and economic outcomes of age at first calving in dairy buffaloes," in *J. Dairy Sci.*, vol. TBC, no. TBC, pp. TBC, 2025, doi: 10.3168/jds.2025-27467.

M. Santinello, G. Neglia, *et al.*, "The role of age at first calving in shaping production and reproductive outcomes in Italian buffaloes," in *J. Dairy Sci.*, vol. 108, no. 7, pp. 7235–7247, Jul. 2025, doi: 10.3168/jds.2025-26369.

Predict Subclinical Mastitis in Italian Buffaloes

Case Study: 6 herds in Southern Italy involving 1,038 Mediterranean buffaloes.

Data:

- **Timeframe:** from August 2019 to February 2021.
- **Milk Data:** Monthly test-day yield, composition (fat, protein, etc.), SCC, DSCC, etc.
- **Animal Info:** Parity, stage of lactation (DIM), calving date.
- **Weather Data:** NASA POWER daily averages (Temp, Humidity, UV, etc.).

Methodology: Compared **GLM, SVM, Random Forest,** and **Neural Network** models.

Performance & Metrics:

- Best Model: **Neural Network (NN)** using Animal ID split*.
- Accuracy = 76.2%.
- Area Under Curve (AUC) = 84.0%.

* To prevent overfitting, two approaches used for splitting data, first splitting by records, second splitting by animal id

G. Neglia *et al.*, "Exploiting machine learning methods with monthly routine milk recording data and climatic information to predict subclinical mastitis in Italian Mediterranean buffaloes," in *J. Dairy Sci.*, vol. 106, no. 3, pp. 1942–1952, Mar. 2023, doi: 10.3168/jds.2022-22292.

Literature Review

Politecnico di Torino

1859

Politecnico di Torino

mohammad.sheikh@studenti.polito.it

Hierarchy in time series (HTS)

Hierarchical time series, organize **time series data** in a nested structure where **higher-level** series aggregate naturally from **lower-level** series.

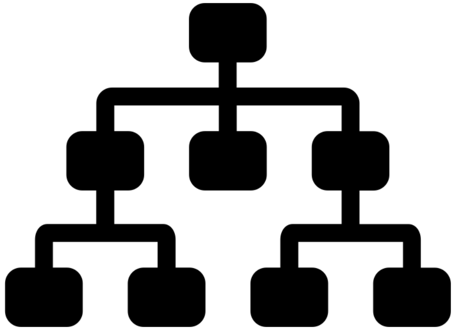
Each level represents a meaningful grouping dimension.

Natural Aggregation: Top-level totals equal the sum of all bottom-level series.

Multiple Levels: Data can be viewed at different levels (national → regional → store level)

Data can be aggregated or disaggregated based on:

- Types
- Features
- geographic division.



R. J. Hyndman and G. Athanasopoulos, *Forecasting: Principles and Practice*, 3rd ed. Melbourne, Australia: OTexts, 2021. [Online]. Available: https://otexts.com/fpp3/

Hierarchy in time series (HTS)

The value of parent is always equals to the sum of the values of its child nodes.

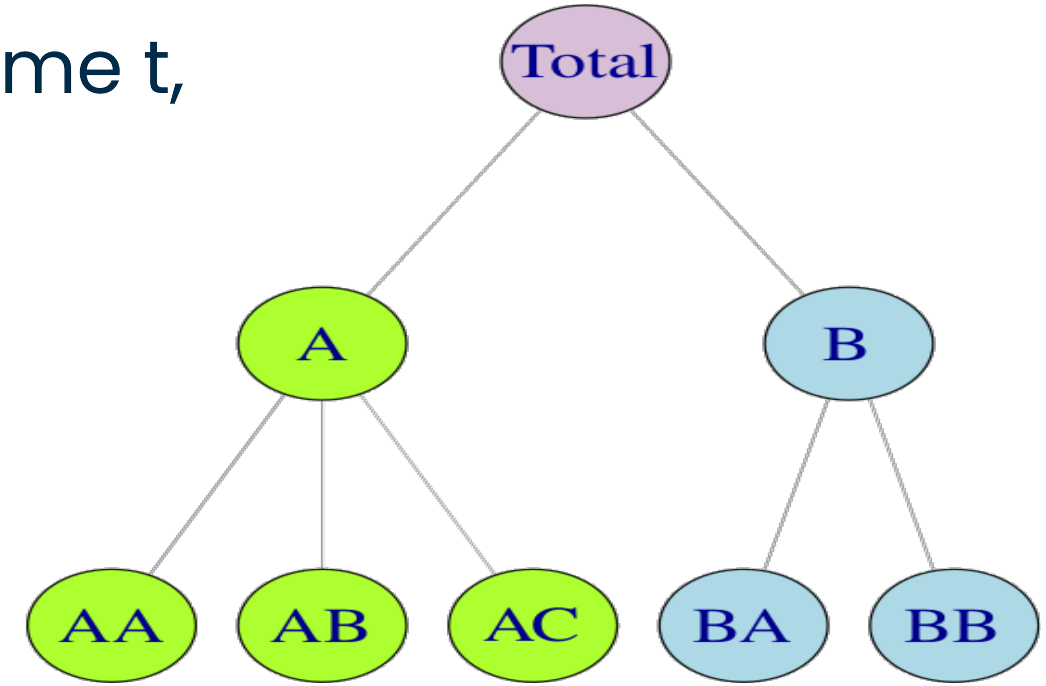
So, if $y_{j,t}$ equals The value of j at time t, then, we have :

$$y_t = y_{\mathrm{A},t} + y_{\mathrm{B},t}.$$

$$y_{\mathrm{A},t} = y_{\mathrm{AA},t} + y_{\mathrm{AB},t} + y_{\mathrm{AC},t}$$

$$y_{\mathrm{B},t} = y_{\mathrm{BA},t} + y_{\mathrm{BB},t}$$

Finally, we can say our data is Coherent.



R. J. Hyndman and G. Athanasopoulos, *Forecasting: Principles and Practice*, 3rd ed. Melbourne, Australia: OTexts, 2021. [Online]. Available: https://otexts.com/fpp3/

Applications on HTS

- **Tourism**

  Forecasting number of tourists going to visit a country divided by **region**, **city**, **season**.

- **supply chain**

  Forecasting sales of products based on **type**, **location**.

- **Energy Consumption**

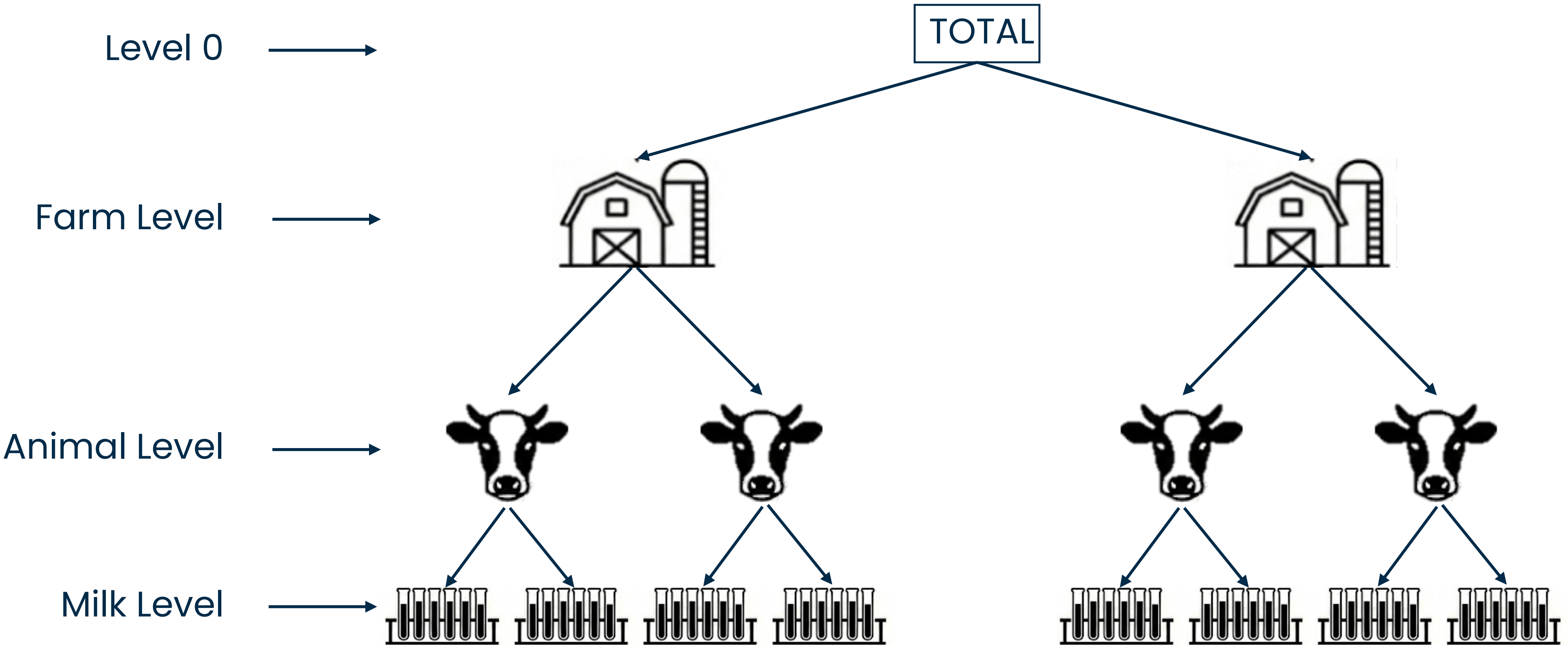
  Forecasting energy consumption based on **region**, **city**, **hour**.

R. J. Hyndman and G. Athanasopoulos, *Forecasting: Principles and Practice*, 3rd ed. Melbourne, Australia: OTexts, 2021. [Online]. Available: https://otexts.com/fpp3/

O. Omri, *et al.*, "On the inventory performance of hierarchical forecasting approaches: case of a vaccine supply chain," J. Oper. Res. Soc., Oct. 2025, doi: 10.1080/01605682.2025.2566323.

M. Kim, *et al.*,"Beginner-friendly review of research on R-based energy forecasting: Insights from text mining," in *Electronics*, vol. 14, no. 17, Art. no. 3513, Sep. 2025, doi: 10.3390/electronics14173513.

Our Case Representation in HTS



Our Case Representation in HTS

Milk Level

Animal_ID	Farm_ID	Date	Age	MILK	DIM	Protein	Fat	SCC
1001	102	25-01-01	4	20 Kg	154	10%	12%	200K/mm
1004	103	25-01-01	6	35 Kg	114	10%	12%	200K/mm
1003	102	25-01-01	5	15 Kg	204	10%	12%	200K/mm
1002	103	25-01-02	7	30 Kg	94	10%	12%	200K/mm
1003	102	25-01-02	5	20 Kg	204	10%	12%	200K/mm
1001	102	25-01-02	4	25 Kg	155	10%	12%	200K/mm
1004	103	25-01-02	4	35 Kg	115	10%	12%	200K/mm
1002	103	25-01-01	7	15 Kg	93	10%	12%	200K/mm

Our Case Representation in HTS

Milk Level

Animal_ID	Farm_ID	Date	Age	MILK	DIM	Protein	Fat	SCC
1001	102	25-01-01	4	20 Kg	154	10%	12%	200K/mm
1004	103	25-01-01	6	35 Kg	114	10%	12%	200K/mm
1003	102	25-01-01	5	15 Kg	204	10%	12%	200K/mm
1002	103	25-01-02	7	30 Kg	94	10%	12%	200K/mm
1003	102	25-01-02	5	20 Kg	204	10%	12%	200K/mm
1001	102	25-01-02	4	25 Kg	155	10%	12%	200K/mm
1004	103	25-01-02	4	35 Kg	115	10%	12%	200K/mm
1002	103	25-01-01	7	15 Kg	93	10%	12%	200K/mm

Our Case Representation in HTS

Animal Level

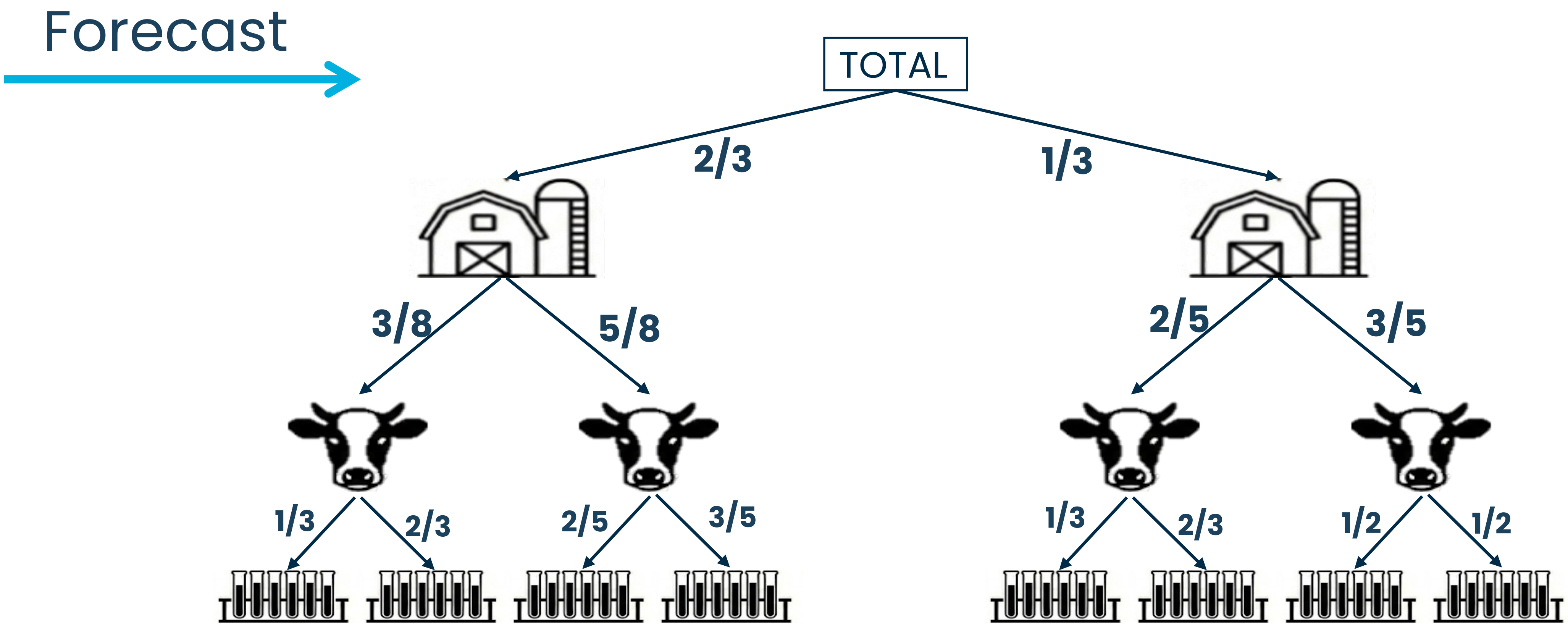
	Animal_ID	Farm_ID	Date	Age	MILK	DIM	Protein	Fat	SCC
Animal 1001	1001	102	25-01-01	4	20 Kg	154	10%	12%	200K/mm
	1001	102	25-01-02	4	25 Kg	155	10%	12%	200K/mm
Animal 1002	1002	103	25-01-02	7	30 Kg	94	10%	12%	200K/mm
	1002	103	25-01-01	7	15 Kg	93	10%	12%	200K/mm
Animal 1003	1003	102	25-01-02	5	20 Kg	204	10%	12%	200K/mm
	1003	102	25-01-01	5	15 Kg	204	10%	12%	200K/mm
Animal 1004	1004	103	25-01-02	4	35 Kg	115	10%	12%	200K/mm
	1004	103	25-01-01	6	35 Kg	114	10%	12%	200K/mm

Our Case Representation in HTS

Farm Level

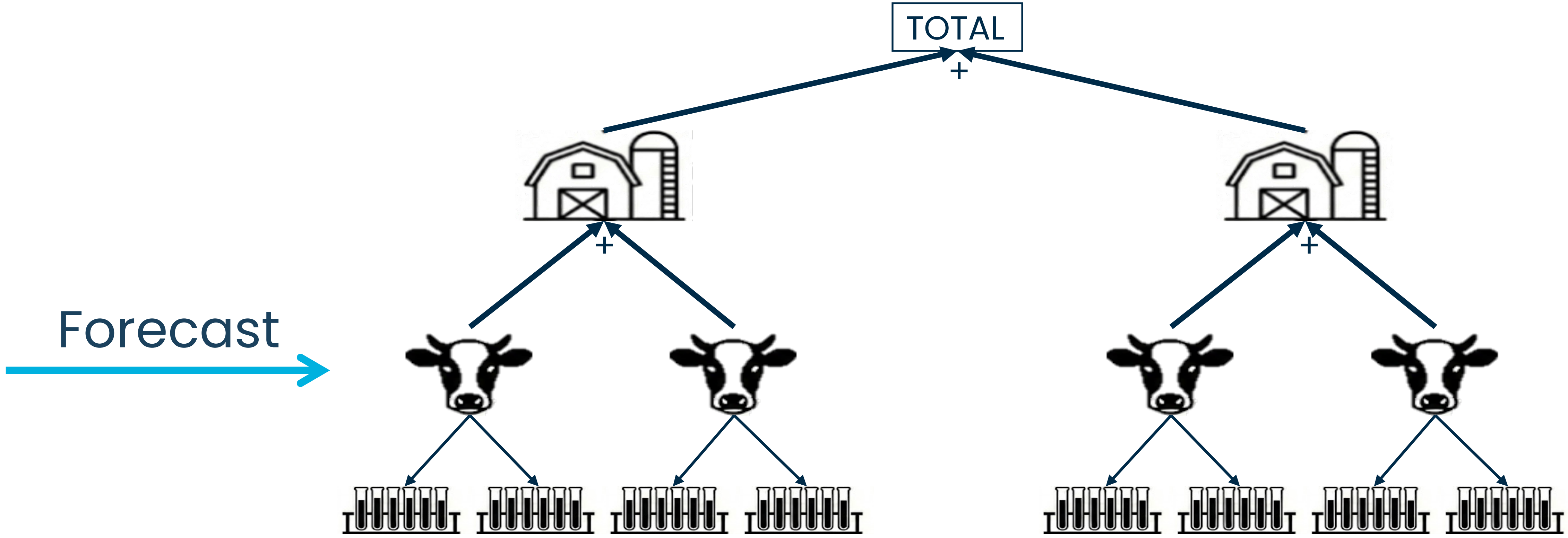
	Animal_ID	Farm_ID	Date	Age	MILK	DIM	Protein	Fat	SCC
Farm 102	1001	102	25-01-01	4	20 Kg	154	10%	12%	200K/mm
	1001	102	25-01-02	4	25 Kg	155	10%	12%	200K/mm
	1003	102	25-01-02	5	20 Kg	204	10%	12%	200K/mm
	1003	102	25-01-01	5	15 Kg	204	10%	12%	200K/mm
Farm 103	1002	103	25-01-02	7	30 Kg	94	10%	12%	200K/mm
	1002	103	25-01-01	7	15 Kg	93	10%	12%	200K/mm
	1004	103	25-01-02	4	35 Kg	115	10%	12%	200K/mm
	1004	103	25-01-01	6	35 Kg	114	10%	12%	200K/mm

Traditional Approaches of Implementation: Top Down



R. J. Hyndman *et al*, "Optimal combination forecasts for hierarchical time series," in *Comput. Stat. Data Anal.*, vol. 55, no. 9, pp. 2579–2589, Sep. 2011, doi: 10.1016/j.csda.2011.03.006.

Traditional Approaches of Implementation: Bottom Up



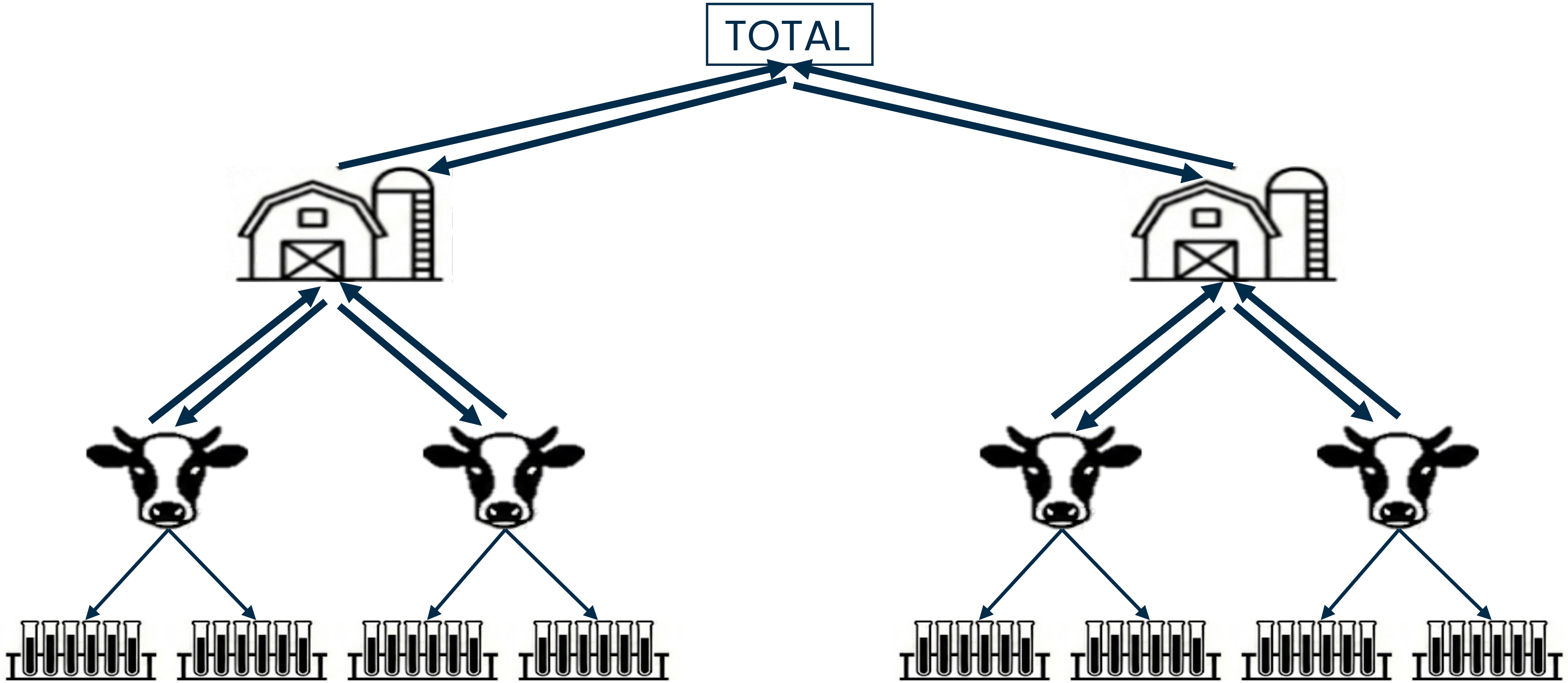
R. J. Hyndman *et al*, "Optimal combination forecasts for hierarchical time series," in *Comput. Stat. Data Anal.*, vol. 55, no. 9, pp. 2579–2589, Sep. 2011, doi: 10.1016/j.csda.2011.03.006.

Traditional Approaches of Implementation: Optimal Combination

Forecast

Forecast

Forecast



S. L. Wickramasuriya, *et al.*, "Optimal forecast reconciliation for hierarchical and grouped time series through trace minimization," in *J. Amer. Stat. Assoc.*, vol. 114, no. 526, pp. 804–819, 2019, doi: 10.1080/01621459.2018.1448825.

Literature Methodology on prediction of HTS

Politecnico di Torino

1859

Politecnico di Torino

mohammad.sheikh@studenti.polito.it

Conditional Hierarchical Forecasting by *M. Abolghasemi,*

Objective: The goal of this approach is to recognize which hierarchical forecasting (HF) method works best given specific data characteristics.

Methodology:

- **Offline Phase**
  - Labeling.
  - Training Classifier.
- **Online Phase**
  - Predict the best method for new data.
  - Forecast future with selected method.

M. Abolghasemi, *et al*, "Model selection in reconciling hierarchical time series," in *Mach. Learn.*, vol. 111, pp. 739–789, Feb. 2022, doi: 10.1007/s10994-021-06126-z.

Conditional Hierarchical Forecasting by *M. Abolghasemi,*

Methodology:

- **Offline Phase**
  - Labeling.
  - Training Classifier.
- **Online Phase**
  - Predict the best method for new data.
  - Forecast future with selected method.

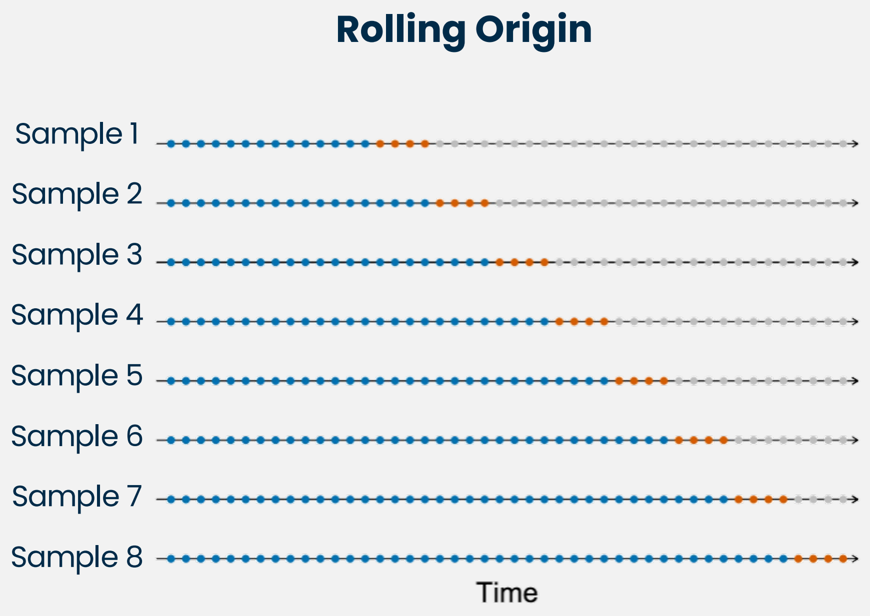
The goal of this phase is to train a machine learning model (classifier) to recognize which hierarchical forecasting (HF) method works best given specific data characteristics.

M. Abolghasemi, *et al,* "Model selection in reconciling hierarchical time series," in *Mach. Learn.*, vol. 111, pp. 739–789, Feb. 2022, doi: 10.1007/s10994-021-06126-z.

Conditional Hierarchical Forecasting by *M. Abolghasemi,*

Methodology:

- **Offline Phase**
  - Labeling.
  - Training Classifier.
- **Online Phase**
  - Predict the best method for new data.
  - Forecast future with selected method.



M. Abolghasemi, *et al,* "Model selection in reconciling hierarchical time series," in *Mach. Learn.*, vol. 111, pp. 739–789, Feb. 2022, doi: 10.1007/s10994-021-06126-z.

Conditional Hierarchical Forecasting by *M. Abolghasemi,*

Methodology:

- **Offline Phase**
  - Labeling.
  - Training Classifier.
- **Online Phase**
  - Predict the best method for new data.
  - Forecast future with selected method.

Parallel Paths

1- Feature Extraction

Algorithm extracts features (Trend, entropy, seasonality, etc.) from each **Sample** and keep it for the next phase.

2- Performance Evaluation

The system generates base forecasts using various standard methods (like Bottom-Up, Top-Down, or Minimum Trace). Then, Calculate the **accuracy** of each **method** and identify the **Winner** as **Label** for corresponding **Sample**

Output

Now, we have some new data for training a classifier to recognize which method works best given time series.

M. Abolghasemi, *et al,* "Model selection in reconciling hierarchical time series," in *Mach. Learn.*, vol. 111, pp. 739–789, Feb. 2022, doi: 10.1007/s10994-021-06126-z.

Conditional Hierarchical Forecasting by *M. Abolghasemi,*

Methodology:

- **Offline Phase**
  - Labeling.
  - Training Classifier.
- **Online Phase**
  - Predict the best method for new data.
  - Forecast future with selected method.

Training the Classifier:

The features (**inputs**) and the most accurate HF method (**labels**) are combined into a classification training set.

A **classification** model (such as XGBoost) is trained to learn the relationship between the time series features and the best performing method .

M. Abolghasemi, *et al*, "Model selection in reconciling hierarchical time series," in *Mach. Learn.*, vol. 111, pp. 739–789, Feb. 2022, doi: 10.1007/s10994-021-06126-z.

Conditional Hierarchical Forecasting by *M. Abolghasemi,*

Methodology:

- **Offline Phase**
  - Labeling.
  - Training Classifier.
- **Online Phase**
  - Predict the best method for new data.
  - Forecast future with selected method.

Predict the best method for new data.

As new data comes in, the algorithm computes the same time series features.

Then, model analyzes these features and predicts/selects which HF method is most likely to be the most accurate for the current data .

M. Abolghasemi, *et al,* "Model selection in reconciling hierarchical time series," in *Mach. Learn.*, vol. 111, pp. 739–789, Feb. 2022, doi: 10.1007/s10994-021-06126-z.

Conditional Hierarchical Forecasting by *M. Abolghasemi,*

Methodology:

- **Offline Phase**
  - Labeling.
  - Training Classifier.
- **Online Phase**
  - Predict the best method for new data.
  - Forecast future with selected method.

Final Forecasting

The system produces base forecasts and reconciles them using *only* the specific HF method selected by the classifier .

M. Abolghasemi, *et al,* "Model selection in reconciling hierarchical time series," in *Mach. Learn.*, vol. 111, pp. 739–789, Feb. 2022, doi: 10.1007/s10994-021-06126-z.

Conditional Hierarchical Forecasting by *M. Abolghasemi,*

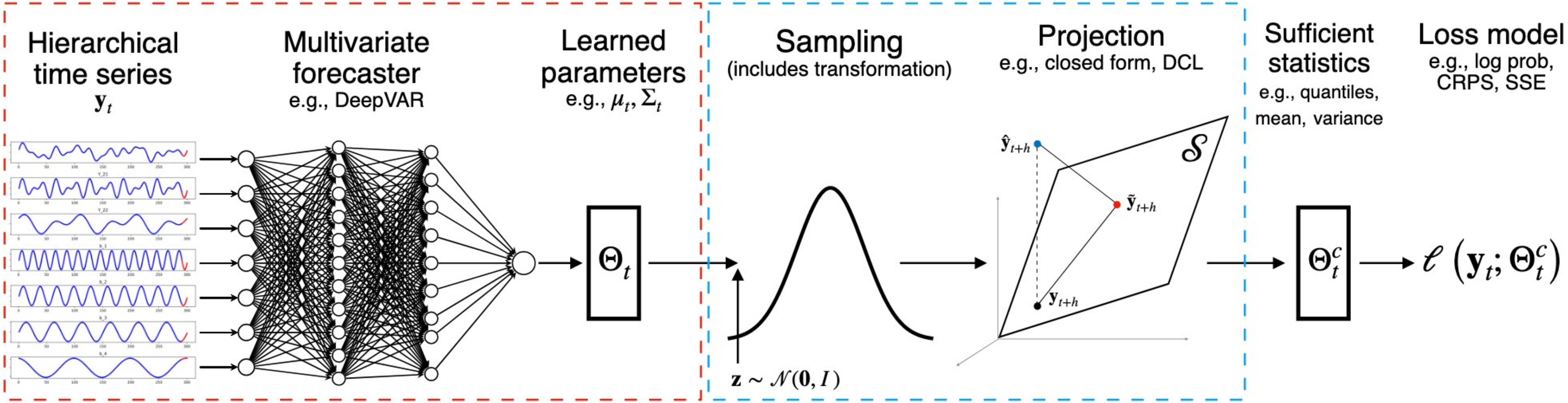
Results

HF Method	Level 0	Level 1	Level 2	Average
MASE				
BU	0.503	0.854	0.856	0.738
TD	**0.435**	0.987	0.907	0.776
COM	0.455	0.844	0.844	0.714
CHF	0.466	**0.820**	**0.828**	**0.705**
RMSSE				
BU	0.583	1.049	1.044	0.892
TD	**0.507**	1.180	1.100	0.929
COM	0.531	1.039	1.035	0.868
CHF	0.534	**1.014**	**1.006**	**0.851**

M. Abolghasemi, *et al,* "Model selection in reconciling hierarchical time series," in *Mach. Learn.*, vol. 111, pp. 739–789, Feb. 2022, doi: 10.1007/s10994-021-06126-z.

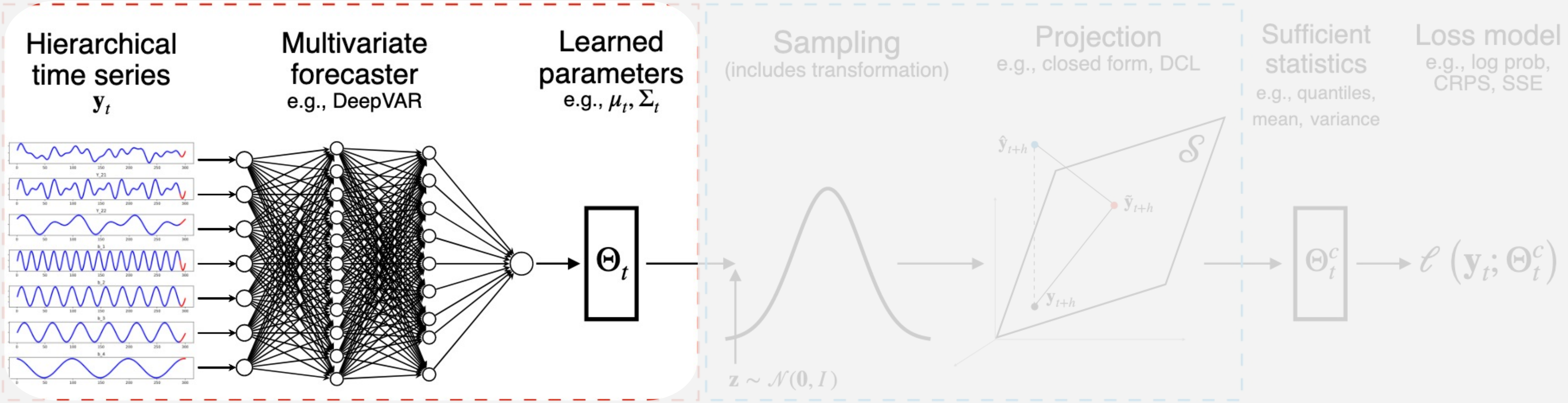
End-to-End Learning of Coherent Probabilistic, by S. Rangapuram

Objective: Learn a single neural network that produces **probabilistic** forecasts for all levels of a hierarchical time series that are **guaranteed to be coherent** without any **post-processing**.



S. Rangapuram, *et al.,* "End-to-end learning of coherent probabilistic forecasts for hierarchical time series," 2021. [Online]. Available: https://www.amazon.science/publications/end-to-end-learning-of-coherent-probabilistic-forecasts-for-hierarchical-time-series

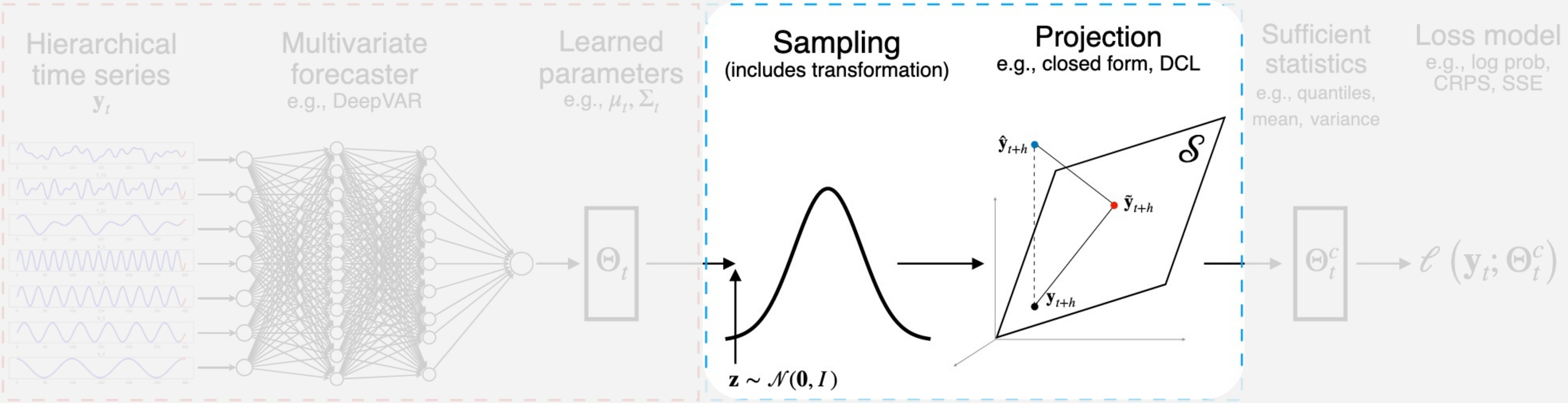
End-to-End Learning of Coherent Probabilistic, by S. Rangapuram



The model uses a **recurrent neural network** (RNN) to learn from **all** time series in the hierarchy **simultaneously**. Then, the model predict the **parameters of data probability distribution** (mean, variances/covariances for all series)

S. Rangapuram, *et al.*, "End-to-end learning of coherent probabilistic forecasts for hierarchical time series," 2021. [Online]. Available: https://www.amazon.science/publications/end-to-end-learning-of-coherent-probabilistic-forecasts-for-hierarchical-time-series

End-to-End Learning of Coherent Probabilistic, by S. Rangapuram

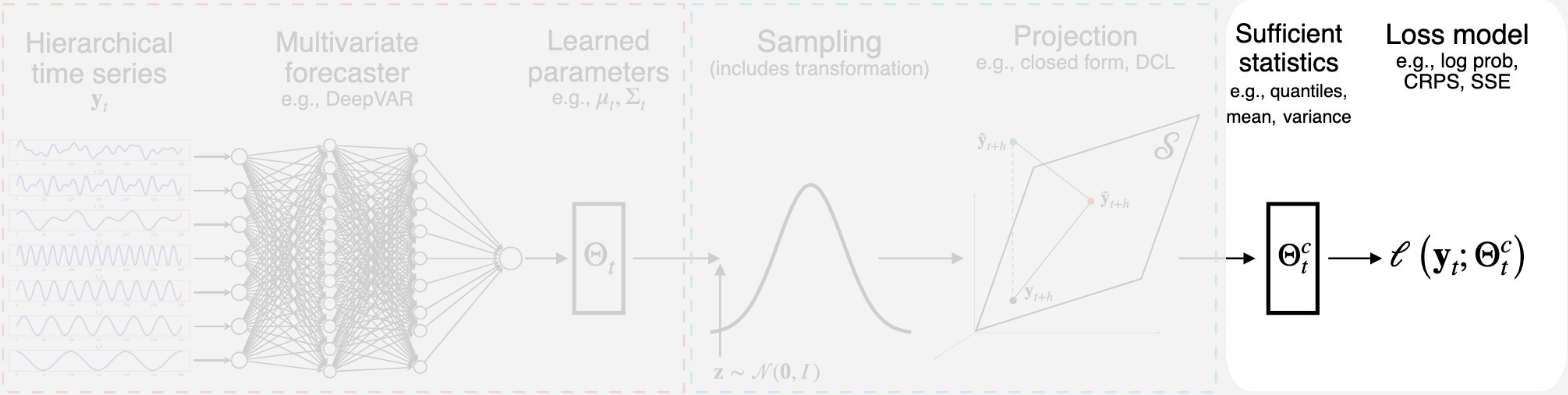


Then, **transforms the** data using **random noise** from normal distribution.

Projection step, where incoherent forecasts become coherent.

S. Rangapuram, *et al.*, "End-to-end learning of coherent probabilistic forecasts for hierarchical time series," 2021. [Online]. Available: https://www.amazon.science/publications/end-to-end-learning-of-coherent-probabilistic-forecasts-for-hierarchical-time-series

End-to-End Learning of Coherent Probabilistic, by S. Rangapuram



Comparing the coherent forecast to the actual observation using **statistics** and **Loss Model.**

Then, **backpropagate:**

Flow back through: Loss ➔ Statistics ➔ Projection ➔ Sampling ➔ Network

Update network weights to make better predictions next time

S. Rangapuram, *et al.*, "End-to-end learning of coherent probabilistic forecasts for hierarchical time series," 2021. [Online]. Available: https://www.amazon.science/publications/end-to-end-learning-of-coherent-probabilistic-forecasts-for-hierarchical-time-series

End-to-End Learning of Coherent Probabilistic, by S. Rangapuram

Results: they use the **continuous ranked probability score** (CRPS) to evaluate the accuracy of their forecast distributions

method	Labour	Traffic	Tourism	Tourism-L	Wiki
ARIMA-NaiveBU	0.0453	0.0808	0.1138	0.1741	0.3772
ETS-NaiveBU	0.0432	0.0665	0.1008	0.1690	0.4673
ARIMA-MinT-shr	0.0467	0.0770	0.1171	0.1609	0.2467
ARIMA-MinT-ols	0.0463	0.1116	0.1195	0.1729	0.2782
ETS-MinT-shr	0.0455	0.0963	0.1013	0.1627	0.3622
ETS-MinT-ols	0.0459	0.1110	0.1002	0.1668	0.2702
ARIMA-ERM	0.0399	0.0466	0.5887	0.5635	0.2206
ETS-ERM	0.0456	0.1027	2.3755	0.5080	0.2217
PERMBU-MINT	0.0393±0.0002	0.0677±0.0061	**0.0771±0.0001**	—	0.2812±0.0240
Hier-E2E (Ours)	**0.0340±0.0088**	**0.0376±0.0060**	0.0834±0.0052	**0.1520±0.0032**	**0.2038±0.0110**

CRPS numbers (lower is better) averaged over 5 runs.

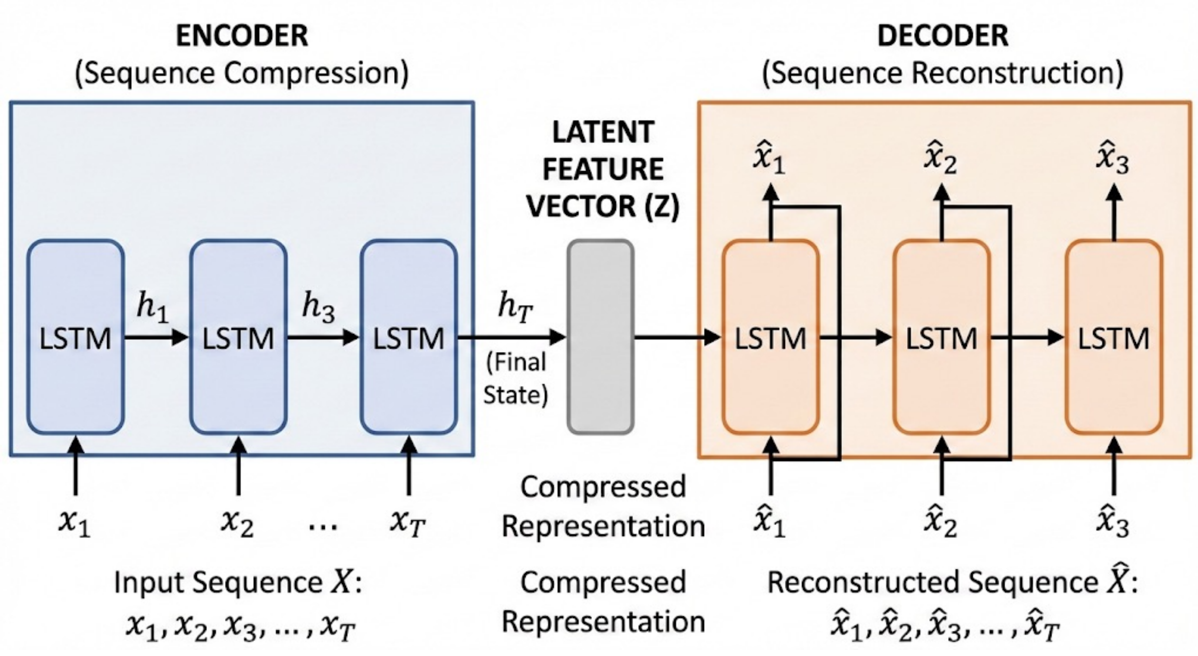
S. Rangapuram, *et al.*, "End-to-end learning of coherent probabilistic forecasts for hierarchical time series," 2021. [Online]. Available: https://www.amazon.science/publications/end-to-end-learning-of-coherent-probabilistic-forecasts-for-hierarchical-time-series

Deep LSTM-Based Transfer Learning Approach, by Sagheer

DLSTM-AE: They developed a Deep Long Short-Term Memory (DLSTM) in an auto-encoder (AE)

Approach:
They are using an **Auto Encoder** (AE) Architecture contains an **encoder** that by getting the sequential data as input, try to learn some **latent features**. And in the **decoder** part try to generate sequential data from the **latent features** found in **encoder**.

For each part of **AE** they use a **LSTM** model (Long short-term Memory) that is an **RNN** model to learn sequent of data.

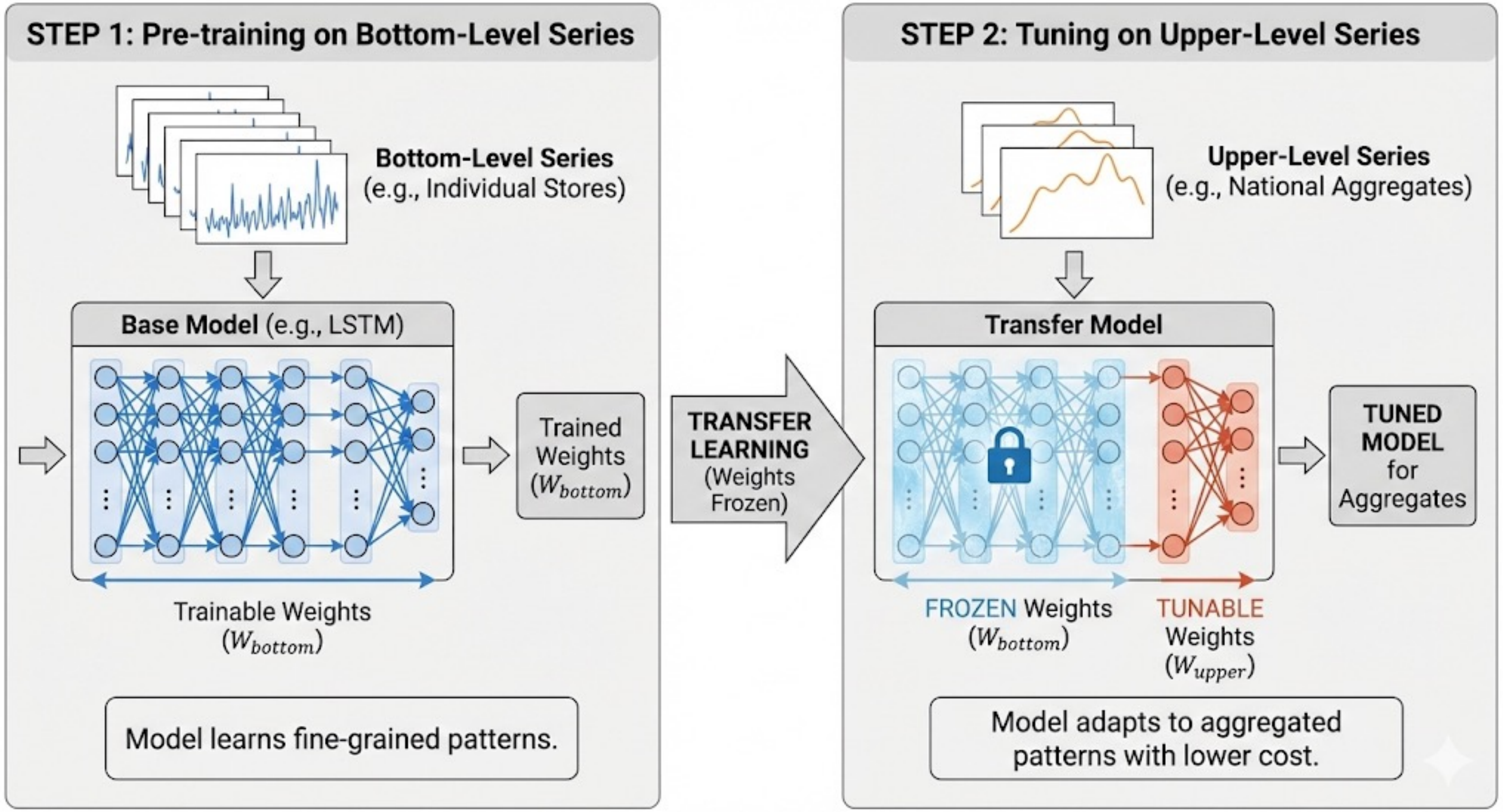


A. Sagheer, *et al.*, "Deep LSTM-based transfer learning approach for coherent forecasts in hierarchical time series," in *Sensors*, vol. 21, no. 13, Art. no. 4379, Jun. 2021, doi: 10.3390/s21134379.

Deep LSTM-Based Transfer Learning Approach, by Sagheer

Transfer Learning:

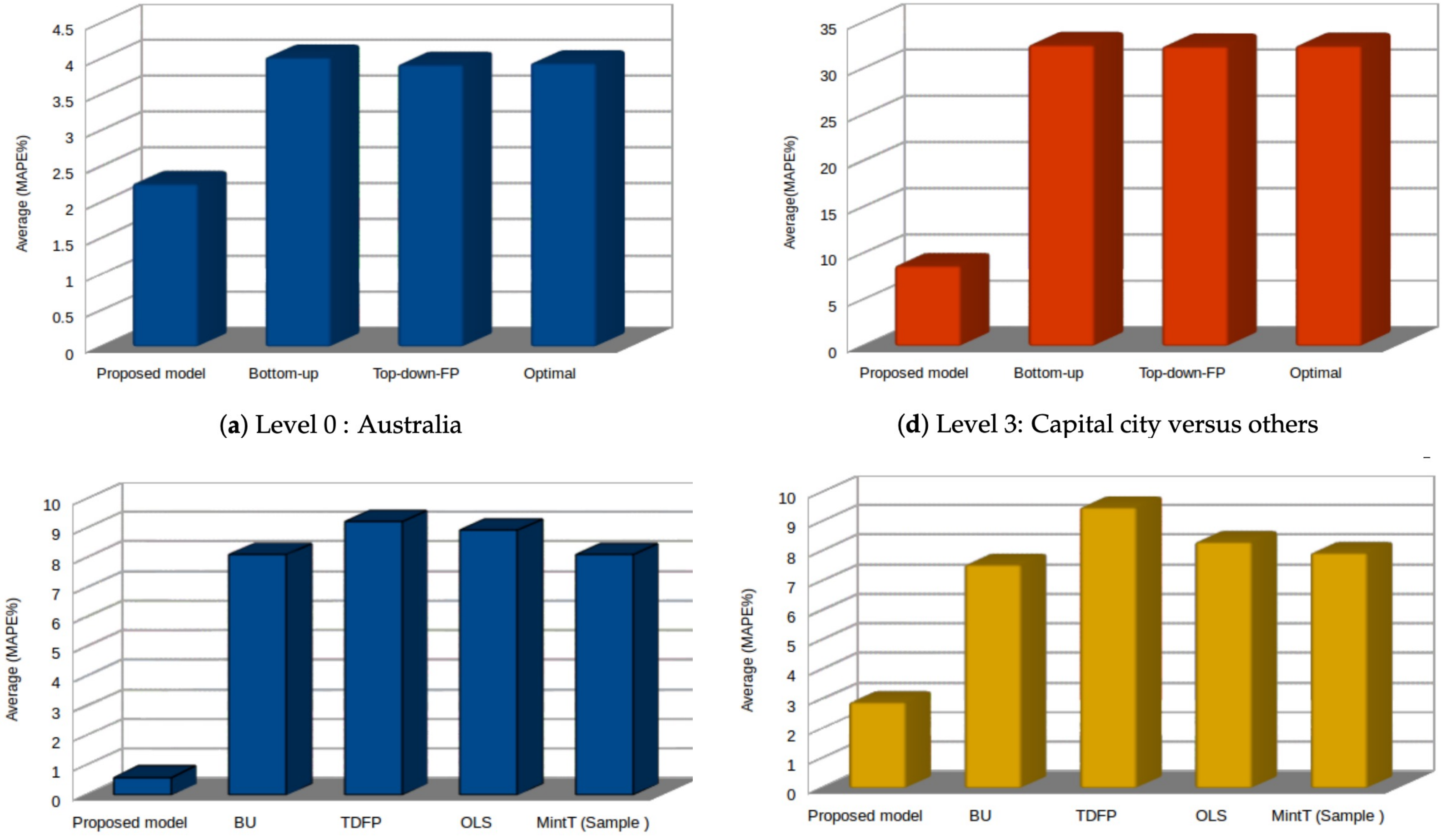
On the other hand, they are using Transfer learning to reduce the computational cost, first they train the model using just bottom level and then the weights of trained model will be frozen and the model tuned on the upper levels for better forecasts.



A. Sagheer, *et al.*, "Deep LSTM-based transfer learning approach for coherent forecasts in hierarchical time series," in *Sensors*, vol. 21, no. 13, Art. no. 4379, Jun. 2021, doi: 10.3390/s21134379.

Deep LSTM-Based Transfer Learning Approach, by Sagheer

Results



(**a**) Level 0 : Australia

(**d**) Level 3: Capital city versus others

(**a**) Level 0 : power generation in Brazil

(**c**) Level 2: total power generation by 14 sources

A. Sagheer, *et al.*, "Deep LSTM-based transfer learning approach for coherent forecasts in hierarchical time series," in *Sensors*, vol. 21, no. 13, Art. no. 4379, Jun. 2021, doi: 10.3390/s21134379.

Neural Network Disaggregation (NND), by P. Mancuso

Objective: Uses deep neural networks (specifically **CNNs**) to directly produce coherent forecasts across hierarchical time series, rather than using traditional post-processing reconciliation methods.

Dataset:
1 - five years Italian grocery store, (3 levels)
2 - electricity demand in Switzerland (2 levels)
3 - Walmart data (M5) (4 levels)
Limitation: a few years of observations for daily time series are needed

They implemented 2 models:

NND1: Direct approach
NND2: Step-by-step approach

P. Mancuso *et al.*,"A machine learning approach for forecasting hierarchical time series," in *Expert Syst. Appl.*, vol. 182, Art. no. 115102, Nov. 2021, doi: 10.1016/j.eswa.2021.115102.

Neural Network Disaggregation (NND), by P. Mancuso

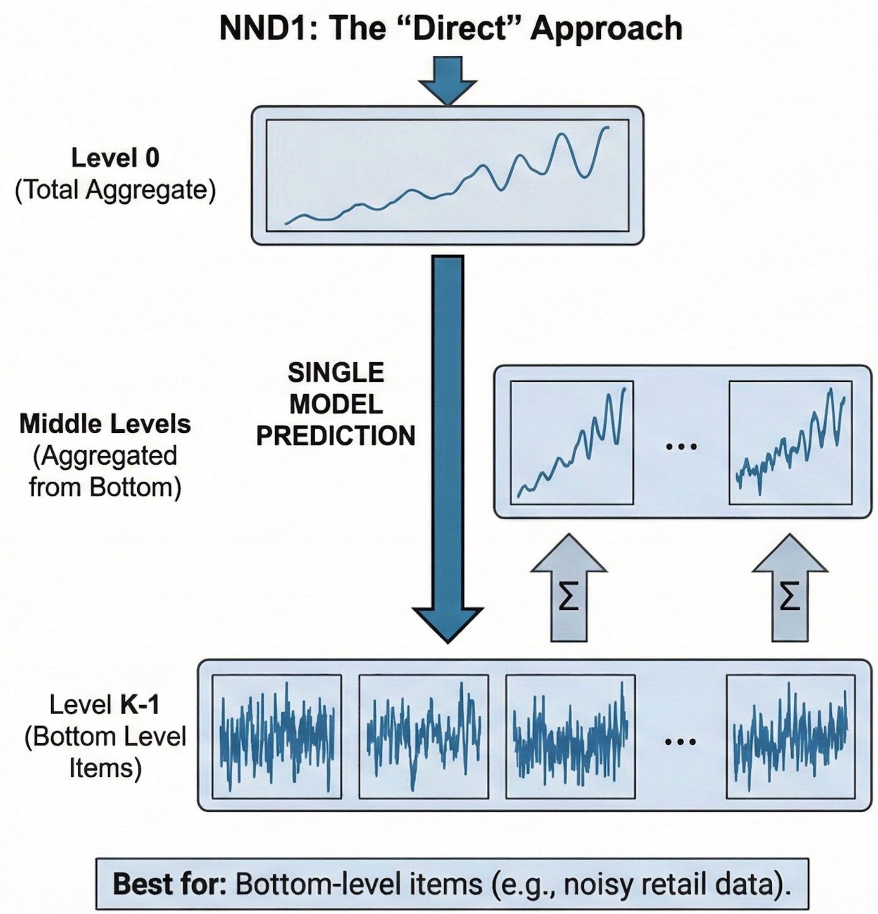
NND1: The "Direct" Approach

The Concept: This method jumps directly from the **very top** of the hierarchy to the **very bottom.**

How it works: A single model is trained to look at the total aggregate data (**Level 0**) and immediately predict the most detailed items at the bottom level (**Level K-1**).

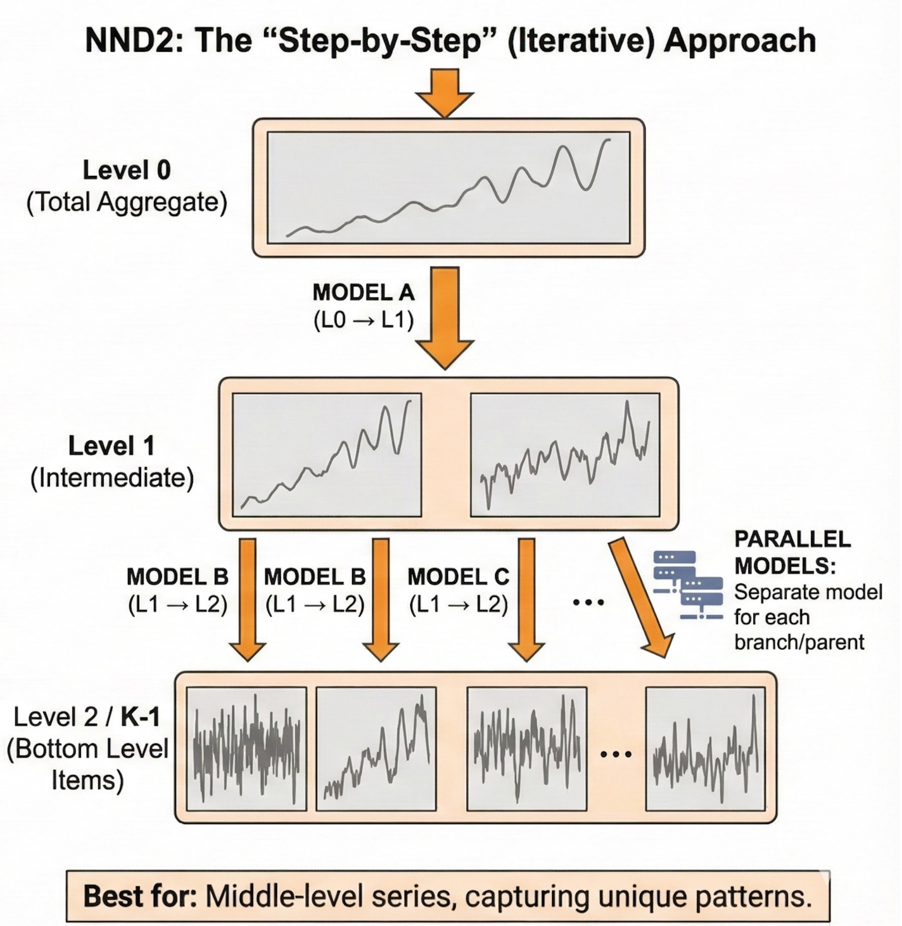
Final Step: Once the **bottom-level** items are predicted, you simply sum them back up to fill in the **middle levels** of the hierarchy.

Best for: This strategy was found to be most effective for the **bottom-level** items.



P. Mancuso *et al.*,"A machine learning approach for forecasting hierarchical time series," in *Expert Syst. Appl.*, vol. 182, Art. no. 115102, Nov. 2021, doi: 10.1016/j.eswa.2021.115102.

Neural Network Disaggregation (NND), by P. Mancuso



NND2: The "Step-by-Step" (Iterative) Approach

The Concept: This method follows the natural structure of the hierarchy **level by level**.

How it works: Instead of one big jump, **separate** models are trained for every single level in the hierarchy. For example, the model first predicts **Level 1 from Level 0**, then **Level 2 from Level 1**, and so on.

Key Difference: It requires training many models in parallel—one for every **parent** series that needs to be broken down.

Best for: This strategy is more flexible and captures the unique patterns of middle levels better.

P. Mancuso *et al.*,"A machine learning approach for forecasting hierarchical time series," in *Expert Syst. Appl.*, vol. 182, Art. no. 115102, Nov. 2021, doi: 10.1016/j.eswa.2021.115102.

Methodology

Politecnico di Torino

Preprocessing

- Removing Outliers and null values
  - Time Series Decomposition. [8]
  - Statical methods like normal distribution.
- Integrating climate data. [3,5]
- THI Composite
  - Avg of 2 days before and after [3]
- Lag Features
  - Captures the effect of past days

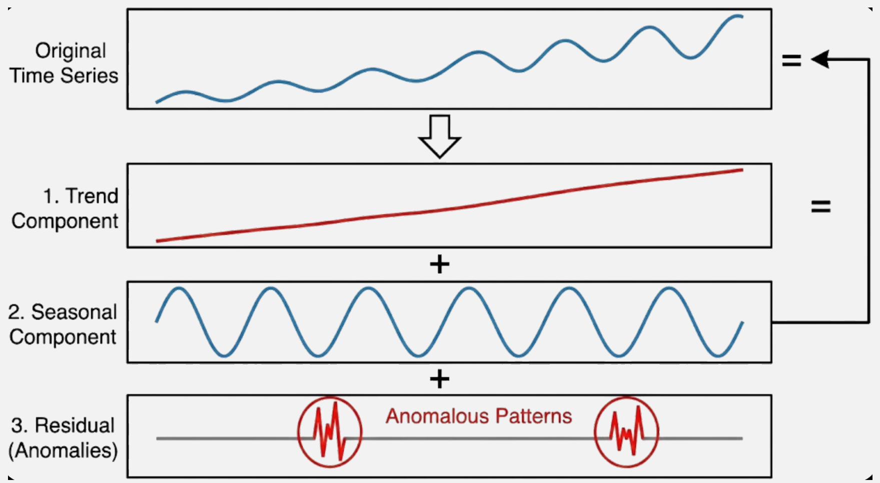
mohammad.sheikh@studenti.polito.it

Preprocessing

- **Removing Outliers and null values**
  - Time Series Decomposition.
  - Statical methods like normal distribution.
- **Integrating climate data.**
- **THI Composite**
  - Avg of 2 days before and after [3]
- **Lag Features**
  - Captures the effect of past days

Time Series Decomposition

Separates the time series into trend, seasonal, and residual components to identify anomalous patterns that deviate significantly from the underlying structure of the data.



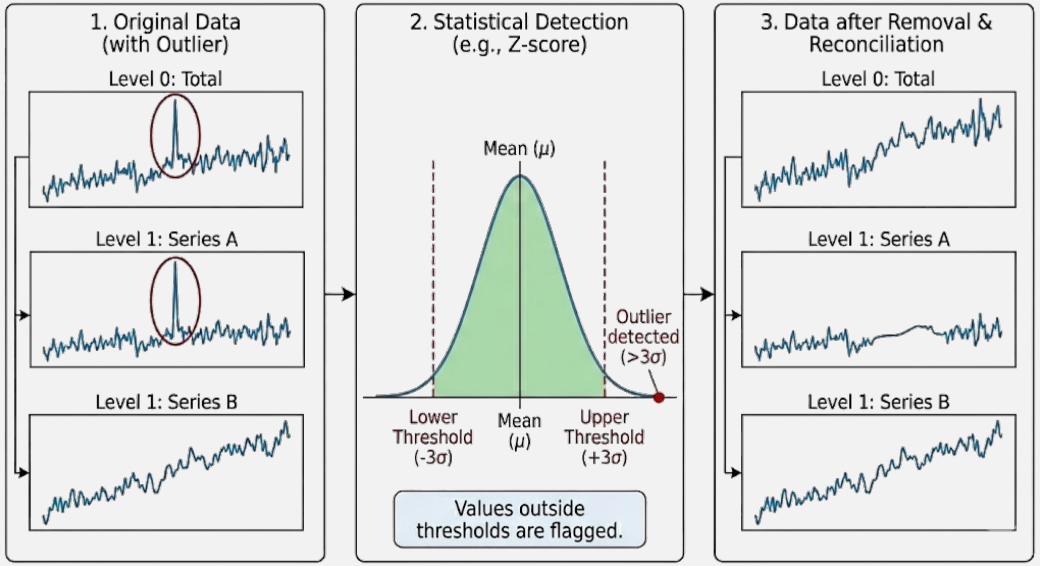
J.-M. Jung, *et al*, "Multi-algorithmic approach for detecting outliers in cattle intake data," in *J. Agric. Food Res.*, vol. 15, Art. no. 101021, Mar. 2024, doi: 10.1016/j.jafr.2024.101021.

Preprocessing

- **Removing Outliers and null values**
  - Time Series Decomposition.
  - Statical methods like normal distribution.
- **Integrating climate data.**
- **THI Composite**
  - Avg of 2 days before and after [3]
- **Lag Features**
  - Captures the effect of past days

Statistical methods like normal distribution

statistical techniques such as **z-scores**, **IQR** or **standard deviation thresholds** to detect and remove values that fall outside expected ranges based on distributional assumptions.



J.-M. Jung, *et al*, "Multi-algorithmic approach for detecting outliers in cattle intake data," in *J. Agric. Food Res.*, vol. 15, Art. no. 101021, Mar. 2024, doi: 10.1016/j.jafr.2024.101021.

Preprocessing

- **Removing Outliers and null values**
  - Time Series Decomposition.
  - Statical methods like normal distribution.
- **Integrating climate data.**
- **THI Composite**
  - Avg of 2 days before and after [3]
- **Lag Features**
  - Captures the effect of past days

Integrating climate data

Incorporates climate variables (**temperature**, **humidity**, **THI** etc.) from **NASA POWER API** as inputs.
using **satellite-based** data instead of physical weather **stations** for consistent coverage across all hierarchical locations.

G. Neglia, *et al.*, "Exploring the sources of variation of electrical conductivity and total and differential somatic cell count in Italian Mediterranean buffaloes," in *J. Dairy Sci.*, vol. 107, no. 1, pp. 508–515, Jan. 2024, doi: 10.3168/jds.2023-23629.

Preprocessing

- **Removing Outliers and null values**
  - Time Series Decomposition.
  - Statical methods like normal distribution.
- **Integrating climate data.**
- **THI Composite**
  - Avg of 2 days before and after
- **Lag Features**
  - Captures the effect of past days

THI Composite

Creates a Temperature-Humidity Index (THI) feature by **averaging** climate conditions over a **5-day** window (2 days before, current day, and 2 days after), smoothing short-term fluctuations and capturing sustained climate stress periods.

$$\text{THI composite} = \frac{\left(\text{THI}_{n-2} + \text{THI}_{n-1} + \text{THI}_{n} + \text{THI}_{n+1} + \text{THI}_{n+2}\right)}{5}$$

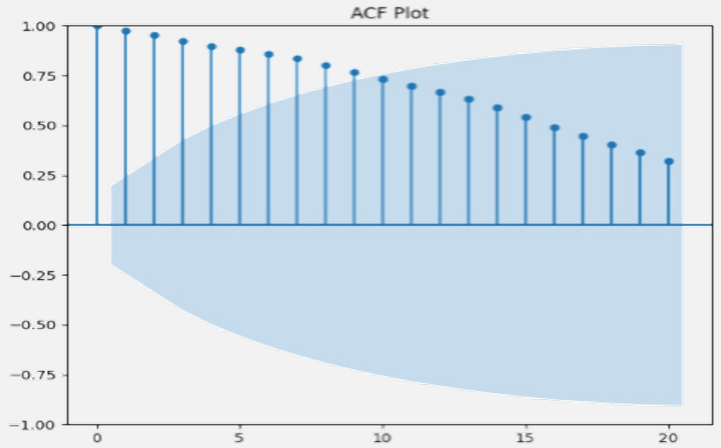
M. F. Hasan, *et al.*,"An effective way to incorporate temperature-humidity index to study effect of heat stress on milk yield by an XGBoost machine learning model," in *J. Dairy Sci.*, vol. 108, no. 12, pp. 13995–14017, Dec. 2025, doi: 10.3168/jds.2025-27193.

Preprocessing

- **Removing Outliers and null values**
  - Time Series Decomposition.
  - Statical methods like normal distribution.
- **Integrating climate data.**
- **THI Composite**
  - Avg of 2 days before and after
- **Lag Features**
  - Captures the effect of past days

Lag Features

Generates lagged variables that include historical values from previous time steps as predictors, allowing the model to learn temporal dependencies and autoregressive patterns where past observations influence future outcomes across the hierarchical structure.



G. Neglia, *et al.*, "Exploring the sources of variation of electrical conductivity and total and differential somatic cell count in Italian Mediterranean buffaloes," in *J. Dairy Sci.*, vol. 107, no. 1, pp. 508–515, Jan. 2024, doi: 10.3168/jds.2023-23629.

Models

References

mohammad.sheikh@studenti.polito.it